



Strathmore
UNIVERSITY

EtherChannel

EtherChannel Operation: Link Aggregation



Strathmore
UNIVERSITY

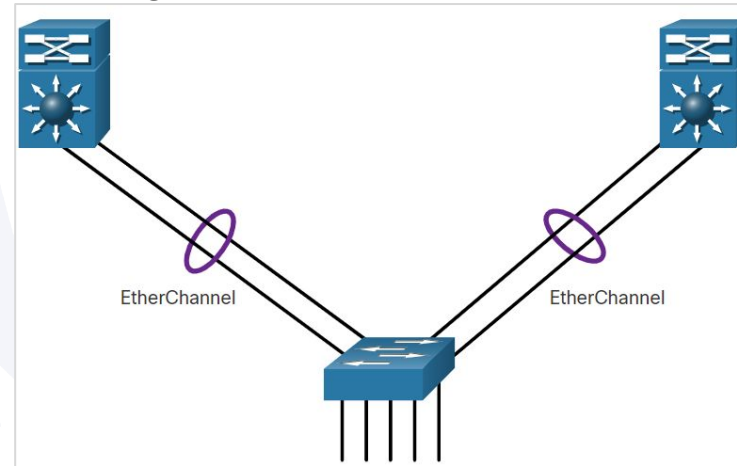
- There are scenarios in which more bandwidth or redundancy between devices is needed than what can be provided by a single link. Multiple links could be connected between devices to increase bandwidth. However, Spanning Tree Protocol (STP), which is enabled on Layer 2 devices like Cisco switches by default, will block redundant links to prevent switching loops.
- A link aggregation technology is needed that allows redundant links between devices that will not be blocked by STP. That technology is known as **EtherChannel**.
- EtherChannel is a link aggregation technology that groups multiple physical Ethernet links together into one single logical link. It is used to provide fault-tolerance, load sharing, increased bandwidth, and redundancy between switches, routers, and servers.
- EtherChannel technology makes it possible to combine the number of physical links between the switches to increase the overall speed of switch-to-switch communication.

EtherChannel Operation: EtherChannel



Strathmore
UNIVERSITY

- EtherChannel technology was originally developed by Cisco as a LAN switch-to-switch technique of grouping several Fast Ethernet or Gigabit Ethernet ports into one logical channel.
- When an EtherChannel is configured, the resulting virtual interface is called a port channel. The physical interfaces are bundled together into a port channel interface, as shown in the figure.



EtherChannel Operation: Link Aggregation



- EtherChannel is a technology that lets you bundle multiple physical links into a single logical link
- Fig 1 shows atypical topology. Consider that H1 sends 800 Mbit of traffic destined for H3 and H2 sends 600 Mbit of traffic destined for H4. The link between the switches will be a bottleneck. We require $800 + 600 = 1400$ Mbit, but we only have a 1000 Mbit link.
- There are two solutions to this problem:
 - Replace the link between the switches with something with a higher bandwidth, perhaps a 10-Gigabit link.
 - Add multiple links and bundle them into an EtherChannel.

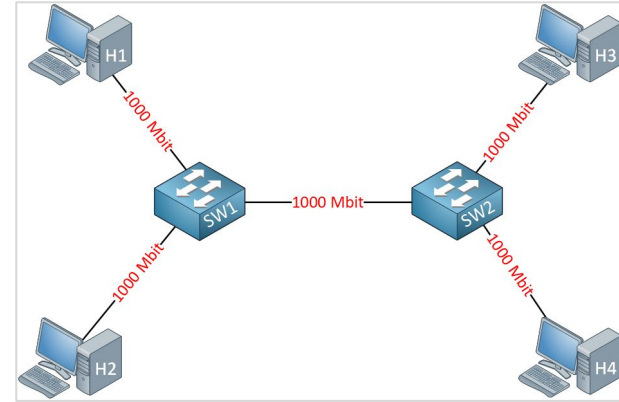


Fig 1

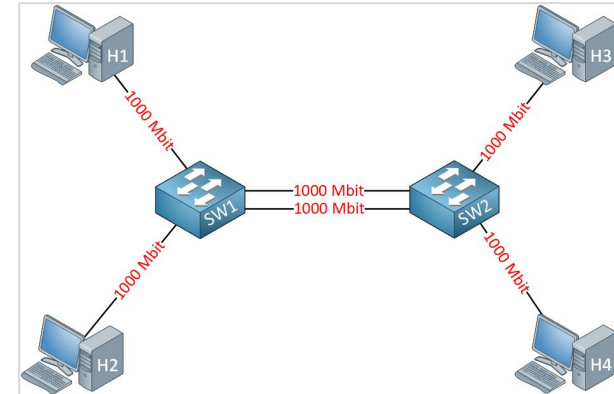


Fig 2

EtherChannel Operation: Link Aggregation



- On Fig 1, An extra link is added. The problem with this setup is that we have a loop between the switches, so spanning tree would block one out of two links. EtherChannel solves this by creating a single virtual link out of these physical links:
- Spanning tree sees one logical link, hence no loops. EtherChannel will do load balancing among the different links and takes care of redundancy. If one of the links fails, it will keep working and use the remaining links.
- Note that the physical links remain the limiting factor. A single traffic flow won't be able to exceed > 1000 Mbit (single Gigabit link). An EtherChannel is the equivalent of adding more lanes to a highway. The bandwidth increases, but the speed limit doesn't change.
- You can assign up to 16 physical interfaces to an EtherChannel, but only eight interfaces will be active at a time.

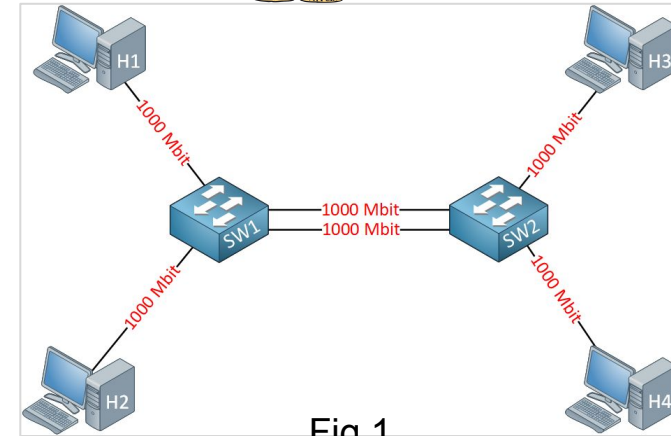


Fig 1

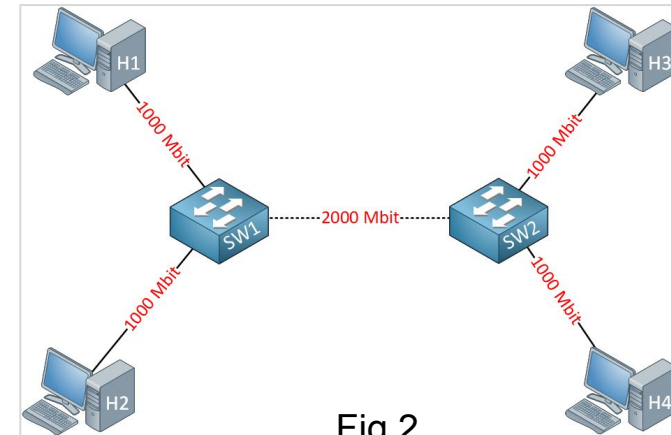


Fig 2

EtherChannel Options



Strathmore
UNIVERSITY

- If you want to configure an EtherChannel then we have three options:
 - Port Aggregation Protocol (PAgP) (Cisco proprietary)
 - Link Aggregation Control Protocol (LACP) (IEEE standard)
 - Manual(**Note:** It is also possible to configure a static or unconditional EtherChannel without PAgP or LACP.)
- PAgP and LACP are negotiation protocols that dynamically configure an Etherchannel. PAgP is a Cisco proprietary protocol (only used between Cisco devices) while LACP is an IEEE standard which many vendors support.
- If you are going to create an EtherChannel you need to make sure that all interfaces have the same configuration:
 - Duplex.
 - Speed.
 - Native and allowed VLANs.
 - Switchport mode (access or trunk).
- PAgP and LACP will check if the configuration of the interfaces that you use is the same.

EtherChannel Operation: Advantages of EtherChannel



Strathmore
UNIVERSITY

EtherChannel technology has many advantages, including the following:

- Most configuration tasks can be done on the EtherChannel interface instead of on each individual port, ensuring configuration consistency throughout the links.
- EtherChannel relies on existing switch ports. There is no need to upgrade the link to a faster and more expensive connection to have more bandwidth.
- Load balancing takes place between links that are part of the same EtherChannel.
- EtherChannel creates an aggregation that is seen as one logical link. When several EtherChannel bundles exist between two switches, STP may block one of the bundles to prevent switching loops. When STP blocks one of the redundant links, it blocks the entire EtherChannel. This blocks all the ports belonging to that EtherChannel link. Where there is only one EtherChannel link, all physical links in the EtherChannel are active because STP sees only one (logical) link.
- EtherChannel provides redundancy because the overall link is seen as one logical connection. Additionally, the loss of one physical link within the channel does not create a change in the topology.

EtherChannel Operation: Implementation Restrictions



Strathmore
UNIVERSITY

EtherChannel has certain implementation restrictions, including the following:

- Interface types cannot be mixed. For example, Fast Ethernet and Gigabit Ethernet cannot be mixed within a single EtherChannel.
- Currently each EtherChannel can consist of up to eight compatibly-configured Ethernet ports. EtherChannel provides full-duplex bandwidth up to 800 Mbps (Fast EtherChannel) or 8 Gbps (Gigabit EtherChannel) between one switch and another switch or host.
- The Cisco Catalyst 2960 Layer 2 switch currently supports up to six EtherChannels.
- The individual EtherChannel group member port configuration must be consistent on both devices. If the physical ports of one side are configured as trunks, the physical ports of the other side must also be configured as trunks within the same native VLAN. Additionally, all ports in each EtherChannel link must be configured as Layer 2 ports.
- Each EtherChannel has a logical port channel interface. A configuration applied to the port channel interface affects all physical interfaces that are assigned to that interface.

EtherChannel Operation: PAgP Operation



Strathmore
UNIVERSITY

- PAgP (pronounced “Pag - P”) is a Cisco-proprietary protocol that aids in the automatic creation of EtherChannel links. When an EtherChannel link is configured using PAgP, PAgP packets are sent between EtherChannel-capable ports to negotiate the forming of a channel.
- When PAgP identifies matched Ethernet links, it groups the links into an EtherChannel. The EtherChannel is then added to the spanning tree as a single port.
- When enabled, PAgP also manages the EtherChannel. PAgP packets are sent every 30 seconds. PAgP checks for configuration consistency and manages link additions and failures between two switches. It ensures that when an EtherChannel is created, all ports have the same type of configuration.
- **Note:** In EtherChannel, it is mandatory that all ports have the same speed, duplex setting, and VLAN information. Any port modification after the creation of the channel also changes all other channel ports.

EtherChannel Operation: PAgP

Operation (Cont.)



Strathmore
UNIVERSITY

- PAgP helps create the EtherChannel link by detecting the configuration of each side and ensuring that links are compatible so that the EtherChannel link can be enabled when needed. The modes for PAgP as follows:
 - **On** - This mode forces the interface to channel without PAgP. Interfaces configured in the on mode do not exchange PAgP packets.
 - **PAgP desirable** - This PAgP mode places an interface in an active negotiating state in which the interface initiates negotiations with other interfaces by sending PAgP packets.
 - **PAgP auto** - This PAgP mode places an interface in a passive negotiating state in which the interface responds to the PAgP packets that it receives but does not initiate PAgP negotiation.

EtherChannel Operation: PAgP Operation (Cont.)



Strathmore
UNIVERSITY

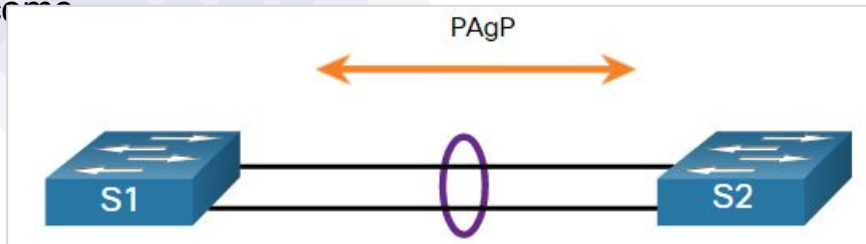
- The modes must be compatible on each side. If one side is configured to be in auto mode, it is placed in a passive state, waiting for the other side to initiate the EtherChannel negotiation.
- If the other side is also set to auto, the negotiation never starts and the EtherChannel does not form.
- If all modes are disabled by using the **no** command, or if no mode is configured, then the EtherChannel is disabled.
- The on mode manually places the interface in an EtherChannel, without any negotiation. It works only if the other side is also set to on. If the other side is set to negotiate parameters through PAgP, no EtherChannel forms, because the side that is set to on mode does not negotiate.
- No negotiation between the two switches means there is no checking to make sure that all the links in the EtherChannel are terminating on the other side, or that there is PAgP compatibility on the other switch.

EtherChannel Operation: PAgP Mode Settings Example



S1	S2	Channel Establishment
On	On	Yes
On	Desirable/Auto	No
Desirable	Desirable	Yes
Desirable	Auto	Yes
Auto	Desirable	Yes
Auto	Auto	No

The table shows the various combination of PAgP modes on S1 and S2 and the resulting channel establishment outcome.



EtherChannel Operation: LACP Operation



Strathmore
UNIVERSITY

- LACP is part of an IEEE specification (802.3ad) that allows several physical ports to be bundled to form a single logical channel. LACP allows a switch to negotiate an automatic bundle by sending LACP packets to the other switch. It performs a function similar to PAgP with Cisco EtherChannel. Because LACP is an IEEE standard, it can be used to facilitate EtherChannels in multivendor environments. On Cisco devices, both protocols are supported.
- LACP provides the same negotiation benefits as PAgP. LACP helps create the EtherChannel link by detecting the configuration of each side and making sure that they are compatible so that the EtherChannel link can be enabled when needed. The modes for LACP are as follows:
 - **On** - This mode forces the interface to channel without LACP. Interfaces configured in the on mode do not exchange LACP packets.
 - **LACP active** - This LACP mode places a port in an active negotiating state. In this state, the port initiates negotiations with other ports by sending LACP packets.
 - **LACP passive** - This LACP mode places a port in a passive negotiating state. In this state, the port responds to the LACP packets that it receives but does not initiate LACP packet negotiation.

EtherChannel Operation: LACP Operation



Strathmore
UNIVERSITY

- LACP is part of an IEEE specification (802.3ad) that allows several physical ports to be bundled to form a single logical channel. LACP allows a switch to negotiate an automatic bundle by sending LACP packets to the other switch. It performs a function similar to PAgP with Cisco EtherChannel. Because LACP is an IEEE standard, it can be used to facilitate EtherChannels in multivendor environments. On Cisco devices, both protocols are supported.
- LACP provides the same negotiation benefits as PAgP. LACP helps create the EtherChannel link by detecting the configuration of each side and making sure that they are compatible so that the EtherChannel link can be enabled when needed. The modes for LACP are as follows:
 - **On** - This mode forces the interface to channel without LACP. Interfaces configured in the on mode do not exchange LACP packets.
 - **LACP active** - This LACP mode places a port in an active negotiating state. In this state, the port initiates negotiations with other ports by sending LACP packets.
 - **LACP passive** - This LACP mode places a port in a passive negotiating state. In this state, the port responds to the LACP packets that it receives but does not initiate LACP packet negotiation.

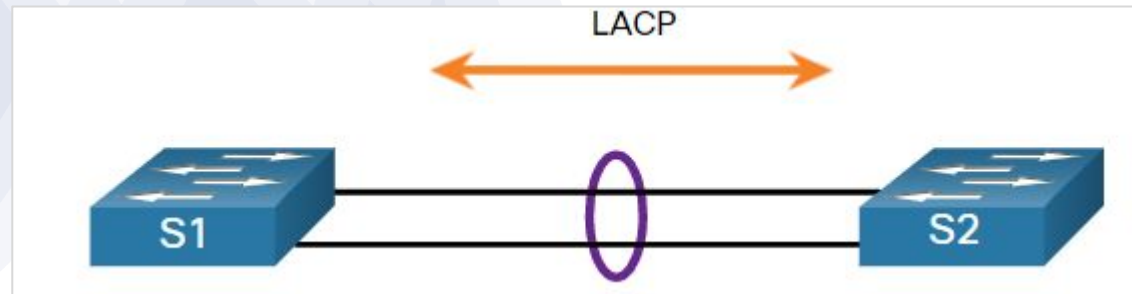
EtherChannel Operation: LACP Mode Settings Example



Strathmore
UNIVERSITY

S1	S2	Channel Establishment
On	On	Yes
On	Active/Passive	No
Active	Active	Yes
Active	Passive	Yes
Passive	Active	Yes
Passive	Passive	No

The table shows the various combination of LACP modes on S1 and S2 and the resulting channel establishment outcome.





Configure EtherChannel

Configure EtherChannel: Configuration Guidelines



Strathmore
UNIVERSITY

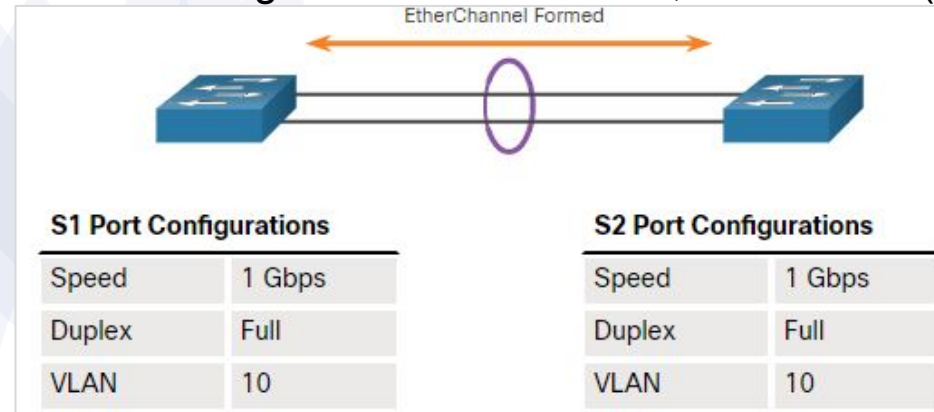
The following guidelines and restrictions are useful for configuring EtherChannel:

- **EtherChannel support** - All Ethernet interfaces must support EtherChannel with no requirement that interfaces be physically contiguous.
- **Speed and duplex** - Configure all interfaces in an EtherChannel to operate at the same speed and in the same duplex mode.
- **VLAN match** - All interfaces in the EtherChannel bundle must be assigned to the same VLAN or be configured as a trunk (shown in the figure).
- **Range of VLANs** - An EtherChannel supports the same allowed range of VLANs on all the interfaces in a trunking EtherChannel. If the allowed range of VLANs is not the same, the interfaces do not form an EtherChannel, even when they are set to **auto** or **desirable** mode.

Configure EtherChannel: Configuration Guidelines (Cont.)



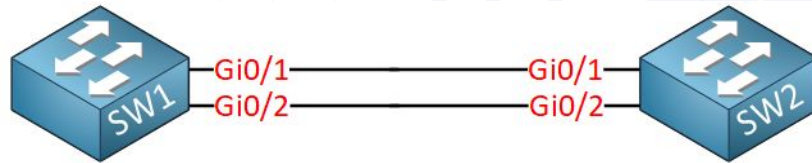
- The figure shows a configuration that would allow an EtherChannel to form between S1 and S2.
- If these settings must be changed, configure them in port channel interface configuration mode. Any configuration that is applied to the port channel interface also affects individual interfaces. However, configurations that are applied to the individual interfaces do not affect the port channel interface. Therefore, making configuration changes to an interface that is part of an EtherChannel link may cause interface compatibility issues.
- The port channel can be configured in access mode, trunk mode (most common), or on a routed port.



Configure EtherChannel: Configuration Guidelines (Cont.)



- Assuming an EtherChannel between two switches, SW1 and SW2 for this demonstration; SW1 and SW2 each have two Gigabit interfaces which need to be bundled into a single logical link.
 - Active/Passive-LACP options
 - Auto/Desirable-PAgP options



```
SW1(config)#interface GigabitEthernet 0/1
```

```
SW1(config-if)#channel-group 1 mode ?
```

active	Enable LACP unconditionally
auto	Enable PAgP only if a PAgP device is detected
desirable	Enable PAgP unconditionally
on	Enable EtherChannel only
passive	Enable LACP only if a LACP device is detected

Configure EtherChannel: PAgP Configuration Example



Strathmore
UNIVERSITY

- Assuming Fig 1 topology, the **channel-group** command on the interface-level is used to set up etherchannel. Any number say 1 can be used to represent the channel group.
- The auto or desirable mode determines whether to actively try to form an EtherChannel or passively wait for the other end to initiate the formation.
- If SW1 is configured for PAgP desirable mode. It will actively ask SW2 to become an EtherChannel this way. We can configure SW2 in either auto or desirable mode. The SW2 configuration uses auto mode:
- With auto mode, SW2 will only respond to incoming PAgP requests. After a few seconds, you'll see the following message on both switches:

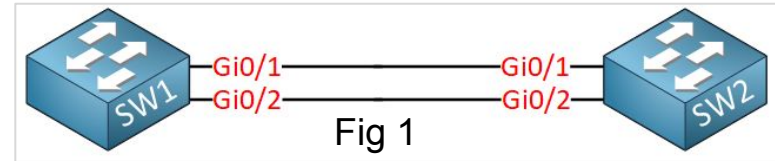
```
SW1(config)#interface range GigabitEthernet 0/1 - 2
```

```
SW1(config-if)#channel-group 1 mode desirable
```

```
SW2(config)#interface range GigabitEthernet 0/1 - 2
```

```
SW2(config-if)#channel-group 1 mode auto
```

```
SW1 %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
```



Configure EtherChannel: PAgP Configuration Example- Verify



Strathmore
UNIVERSITY

The switch automatically creates a new port-channel interface: Fig 1
Port channel aggregated using PAgP. Fig 2

SW1#show etherchannel summary

Flags: D - down P - bundled in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator

M - not in use, minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1

Number of aggregators: 1

Fig 2

Group	Port-channel	Protocol	Ports
1	Po1(SU)	PAgP	Gi0/1(P) Gi0/2(P)

SW1#show interfaces port-channel 1

Port-channel1 is up, line protocol is up (connected)

Hardware is EtherChannel, address is fa16.3e15.5fb2 (bia fa16.3e15.5fb2)

MTU 1500 bytes, BW 2000000 Kbit/sec, DLY 10 usec,

reliability 255/255, txload 1/255, rxload 1/255

Encapsulation ARPA, loopback not set

Keepalive set (10 sec)

ARP type: ARPA, ARP Timeout 04:00:00

Last input 00:03:04, output never, output hang never

Last clearing of "show interface" counters never

Input queue: 0/2000/0/0 (size/max/drops/flushes); Total output drops: 0

Queueing strategy: fifo

Output queue: 0/40 (size/max)

5 minute input rate 0 bits/sec, 0 packets/sec

5 minute output rate 0 bits/sec, 0 packets/sec

12 packets input, 973 bytes, 0 no buffer

Received 0 broadcasts (0 multicasts)

0 runs, 0 giants, 0 throttles

0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored

0 input packets with dribble condition detected

240 packets output, 17496 bytes, 0 underruns

0 output errors, 0 collisions, 0 interface resets

0 unknown protocol drops

0 babbles, 0 late collision, 0 deferred

0 lost carrier, 0 no carrier

0 output buffer failures, 0 output buffers swapped out

Fig 1

Configure EtherChannel: PAgP Configuration Example- Verify



Strathmore
UNIVERSITY

*Show switch Etherchannel configs PAgP:
Fig 1*

Show switch interface on Etherchannel

```
SW1#show etherchannel 1 port-channel
```

Port-channels in the group:

Port-channel: Po1

Age of the Port-channel = 0d:00h:10m:16s

Logical slot/port = 2/1 Number of ports = 2

GC = 0x00010001 HotStandBy port = null

Port state = Port-channel Ag-Inuse

Protocol = PAgP

Port security = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
-----+-----+-----+-----+-----				
0	00	gi0/1	Desirable-Sl	0
0	00	gi0/2	Desirable-Sl	0

Time since last port bundled: 0d:00h:00m:07s Gi0/1

Time since last port Un-bundled: 0d:00h:04m:08s Gi0/2

Fig 2

```
SW1#show interfaces GigabitEthernet 0/1 etherchannel
```

Port state = Up Mstr In-Bndl

Channel group = 1 Mode = Desirable-Sl Gcchange = 0

Port-channel = Po1 GC = 0x00010001 Pseudo port-channel = Po1

Port index = 0 Load = 0x00 Protocol = PAgP

Flags: S - Device is sending Slow hello. C - Device is in Consistent state.
A - Device is in Auto mode. P - Device learns on physical port.
d - PAgP is down.

Timers: H - Hello timer is running. Q - Quit timer is running.
S - Switching timer is running. I - Interface timer is running.

Local information:

Port	Flags	State	Timers	Hello Interval	Partner Count	PAgP Priority	Learning Method	Group Ifindex
Gi0/1	SC	U6/S7	H	30s	1	128	Any	5001

Partner's information:

Port	Partner Name	Partner Device ID	Partner Port	Partner Age	Partner Flags	Partner Group Cap.
Gi0/1	SW2	0019.569d.5700	Gi0/1	19s	SAC	10001

Age of the port in the current state: 0d:00h:02m:37s

Fig 1

Configure EtherChannel: PAgP Configuration Example-Verify



Strathmore
UNIVERSITY

- At the moment, there's nothing configured under the port-channel interface: Fig 1
- Our two physical interfaces that belong to this EtherChannel only have the *channel-group* command: Fig 2-3
- If you want to make any changes like configuring the EtherChannel as a trunk, you need to do this under the port-channel interface. For example:
- When you do this, the switch automatically adds the commands to the physical interfaces as well: Fig 5-6
- You shouldn't make any changes to the physical interfaces that belong to the EtherChannel. Always use the port-channel interface.

```
SW1#show run interface Port-channel 1
```

```
interface Port-channel1
end
```

FIG 1

```
SW1#show run interface GigabitEthernet 0/1
```

```
interface GigabitEthernet0/1
channel-group 1 mode desirable
end
```

FIG 2

```
SW1#show run interface GigabitEthernet 0/2
```

```
interface GigabitEthernet0/2
channel-group 1 mode desirable
end
```

FIG 3

```
SW1(config)#interface Port-channel 1
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#switchport mode trunk
```

FIG 4

```
SW1#show run interface GigabitEthernet 0/1 SW1#show run interface GigabitEthernet 0/2
```

```
interface GigabitEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
end
```

FIG 5

```
interface GigabitEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
end
```

FIG 6

Configure EtherChannel: LACP Configuration Example



Strathmore
UNIVERSITY

Configuring EtherChannel with LACP requires the following three steps:

- **Step 1.** Specify the interfaces that compose the EtherChannel group using the **interface range** *interface* global configuration mode command. The **range** keyword allows you to select several interfaces and configure them all together.
- **Step 2.** Create the port channel interface with the **channel-group identifier mode active** command in interface range configuration mode. The identifier specifies a channel group number. The **mode active** keywords identify this as an LACP EtherChannel configuration.
- **Step3.** To change Layer 2 settings on the port channel interface, enter port channel interface configuration mode using the **interface port-channel** command, followed by the interface identifier. In the example, S1 is configured with an LACP EtherChannel. The port channel is configured as a trunk interface with the allowed VLANs specified.

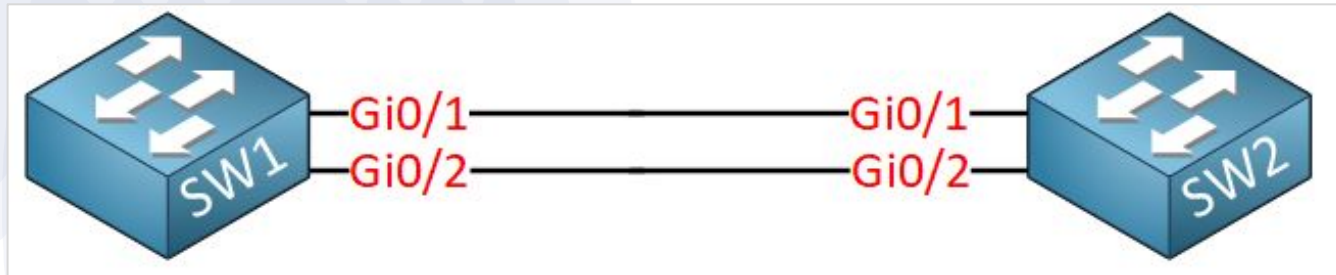
```
S1(config)# interface range FastEthernet 0/1 - 2
S1(config-if-range)# channel-group 1 mode active
Creating a port-channel interface Port-channel 1
S1(config-if-range)# exit
S1(config-if)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 1,2,20
```

Configure EtherChannel: LACP Configuration Example



- LACP is similar to PAgP but uses different terminology:
 - **Active:** The interface will actively ask the other side to become an EtherChannel.
 - **Passive:** The interface waits passively for the other side to ask to become an EtherChannel.
- At least one switch should use active mode. When both switches use passive mode, nothing will happen.

	Active	Passive
Active	Yes	Yes
Passive	Yes	No



Configure EtherChannel: LACP Configuration Example



Strathmore
UNIVERSITY

- Assuming Fig 1 topology, let's configure SW1(Fig 2) to use active mode and SW2 in passive mode (Fig 3)



```
SW1(config-if)#interface range GigabitEthernet 0/1 - 2
```

```
SW1(config-if)#channel-group 1 mode active
```

Fig 2

```
SW2(config)#interface range GigabitEthernet 0/1 - 2
```

```
SW2(config-if)#channel-group 1 mode passive
```

Fig 3

Let's verify our work. Fig 4

```
SW1#show etherchannel summary
```

Flags: D - down P - bundled in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use N - not in use, no aggregation

f - failed to allocate aggregator

M - not in use, minimum links not met

m - not in use, port not aggregated due to minimum links not met

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
-------	--------------	----------	-------

1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(P)
---	---------	------	-------------------

Fig 4

Configure EtherChannel: LACP Configuration Example



Strathmore
UNIVERSITY

- Show Etherchannel configuration options: Fig1-2

```
SW1#show etherchannel 1 port-channel
```

Port-channels in the group:

Port-channel: Po1 (Primary Aggregator)

Age of the Port-channel = 0d:00h:01m:43s

Logical slot/port = 16/0 Number of ports = 2

HotStandBy port = null

Port state = Port-channel Ag-Inuse

Protocol = LACP

Port security = Disabled

Load share deferral = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/1	Active	0
0	00	Gi0/2	Active	0

Fig1

Time since last port bundled: 0d:00h:01m:04s Gi0/2

```
SW1#show interfaces GigabitEthernet 0/1 etherchannel
```

Port state = Up Mstr Assoc In-Bndl

Channel group = 1 Mode = Active

Gcchange = -

Port-channel = Po1

GC = -

Pseudo port-channel = Po1

Port index = 0

Load = 0x00

Protocol = LACP

Flags: S - Device is sending Slow LACPDUs F - Device is sending fast LACPDUs.

A - Device is in active mode.

P - Device is in passive mode.

Local information:

Port	Flags	State	LACP port Priority	Admin Key	Oper Key	Port Number	Port State
Gi0/1	SA	bndl	32768	0x1	0x1	0x2	0x3D

Partner's information:

Port	Flags	LACP port Priority	Dev ID	Age	Admin key	Oper Key	Port Number	Port State
Gi0/1	SP	32768	5e00.0001.8000	18s	0x0	0x1	0x2	0x3C

Age of the port in the current state: 0d:00h:01m:37s

Fig2

Configure EtherChannel: Manual Configuration Example



Strathmore
UNIVERSITY

- Instead of PAgP or LACP, we can also manually enable the Etherchannel. Let's use the same topology(Fig1) to demonstrate this(Fig2)
- Review the results Fig 3



```
SW1(config)#interface range GigabitEthernet 0/1 - 2
SW1(config-if-range)#channel-group 1 mode on
```

```
SW2(config)#interface range GigabitEthernet 0/1 - 2
SW2(config-if-range)#channel-group 1 mode on
```

Fig 2

```
SW1#show etherchannel summary
```

Flags: D - down P - bundled in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use N - not in use, no aggregation

f - failed to allocate aggregator

M - not in use, minimum links not met

m - not in use, port not aggregated due to minimum links not met

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	-	Gi0/1(P) Gi0/2(P)

Fig 3

Configure EtherChannel: Manual Configuration Example



Strathmore
UNIVERSITY

- Configuration is manual
- we can check one of the two physical interfaces and see the Etherchannel information.

```
SW1#show interfaces GigabitEthernet 0/1 etherchannel
Port state      = Up Mstr In-Bndl
Channel group = 1          Mode = On          Gcchange = -
Port-channel   = Po1       GC      = -        Pseudo port-channel = Po1
Port index     = 0         Load = 0x00      Protocol = -

Age of the port in the current state: 0d:00h:01m:41s
```

Fig 1

```
SW1#show etherchannel 1 port-channel
```

Port-channels in the group:

Port-channel: Po1

Age of the Port-channel = 0d:00h:06m:54s

Logical slot/port = 16/0 Number of ports = 2

GC = 0x00000000 HotStandBy port = null

Port state = Port-channel Ag-Inuse

Protocol = -

Port security = Disabled

Load share deferral = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/1	On	0
0	00	Gi0/2	On	0

Time since last port bundled: 0d:00h:01m:21s Gi0/2

Time since last port Un-bundled: 0d:00h:04m:05s Gi0/2

Fig2



Verify and Troubleshoot EtherChannel

Verify and Troubleshoot EtherChannel:

Verify EtherChannel



Strathmore
UNIVERSITY

- As always, when you configure devices in your network, you must verify your configuration. If there are problems, you will also need to be able to troubleshoot and fix them. There are a number of commands to verify an EtherChannel configuration:
 - The **show interfaces port-channel** command displays the general status of the port channel interface.
 - The **show etherchannel summary** command displays one line of information per port channel.
 - The **show etherchannel port-channel** command displays information about a specific port channel interface.
 - The **show interfaces etherchannel** command can provide information about the role of a physical member interface of the EtherChannel.

Verify and Troubleshoot EtherChannel:

Common Issues with EtherChannel Configurations



Strathmore
UNIVERSITY

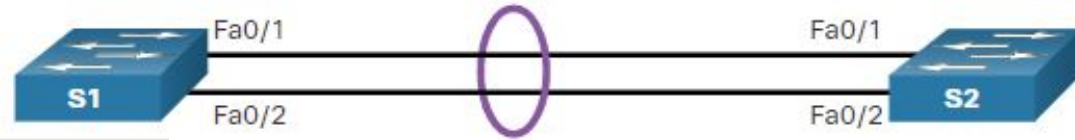
- All interfaces within an EtherChannel must have the same configuration of speed and duplex mode, native and allowed VLANs on trunks, and access VLAN on access ports. Ensuring these configurations will significantly reduce network problems related to EtherChannel. Common EtherChannel issues include the following:
 - Assigned ports in the EtherChannel are not part of the same VLAN, or not configured as trunks. Ports with different native VLANs cannot form an EtherChannel.
 - Trunking was configured on some of the ports that make up the EtherChannel, but not all of them. It is not recommended that you configure trunking mode on individual ports that make up the EtherChannel. When configuring a trunk on an EtherChannel, verify the trunking mode on the EtherChannel.
 - If the allowed range of VLANs is not the same, the ports do not form an EtherChannel even when PAgP is set to the **auto** or **desirable** mode.
 - The dynamic negotiation options for PAgP and LACP are not compatibly configured on both ends of the EtherChannel.

Verify and Troubleshoot EtherChannel: Troubleshoot EtherChannel Example



Strathmore
UNIVERSITY

- In the figure, interfaces Fa0/1 and Fa0/2 on switches S1 and S2 are connected with an EtherChannel. However, the EtherChannel is not operational.



```
S1# show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       N - not in use, no aggregation
       f - failed to allocate aggregator
       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SD)        -           Fa0/1(D)   Fa0/2(D)
```

- Step 1. View the EtherChannel Summary Information:** The output of the **show etherchannel summary** command indicates that the EtherChannel is down.

Verify and Troubleshoot EtherChannel: Troubleshoot EtherChannel Example (Cont.)



Strathmore
UNIVERSITY

Step 3

```
S1(config)# no interface port-channel 1
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# interface range fa0/1 - 2
S1(config-if-range)# channel-group 1 mode desirable
S1(config-if-range)# no shutdown
S1(config-if-range)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1#
```

•**Step 3: Correct the Misconfiguration:** To correct the issue, the PAgP mode on the EtherChannel is changed to desirable.

•**Note:** EtherChannel and STP must interoperate. For this reason, the order in which EtherChannel-related commands are entered is important, which is why you see interface Port-Channel 1 removed and then re-added with the **channel-group** command, as opposed to directly changed. If one tries to change the configuration directly, STP errors cause the associated ports to go into blocking or errdisabled state.

•**Step 2. View Port Channel Configuration:** In the **show run | begin interface port-channel** output, more detailed output indicates that there are incompatible PAgP modes configured on S1 and S2.

```
S1# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode on
!=====
S2# show run | begin interface port-channel
interface Port-channel1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
!
interface FastEthernet0/1
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
!
interface FastEthernet0/2
  switchport trunk allowed vlan 1,2,20
  switchport mode trunk
  channel-group 1 mode desirable
```

Step 2

Verify and Troubleshoot EtherChannel: Troubleshoot EtherChannel Example (Cont.)



```
S1# show etherchannel summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

```
Group  Port-channel  Protocol  Ports
```

```
-----+-----+-----+-----+-----+
1      Po1(SU)      PAgP      Fa0/1(P)  Fa0/2(P)
```

- **Step 4. Verify EtherChannel is Operational:** The EtherChannel is now active as verified by the output of the **show etherchannel summary** command.

EtherChannel Load Balancing



Strathmore
UNIVERSITY

- EtherChannel supports load balancing. However, it does not mean that the traffic is distributed equally among the links. The traffic that goes through the port-channel interface is not forwarded on a round-robin basis. Instead, EtherChannel load balancing uses a hash algorithm to forward packets.
- The calculated load balancing hash determines which physical interface will be used to forward the packet. The load balancing method can be configured using the '**port-channel load-balance <hash>**' global configuration command and have the following hash options or keywords which are based on the source and destination IP address, MAC address, and TCP/UDP ports:
- Use the **show etherchannel load-balance** command to see what the default configuration is. As you can see, this switch load balances based on the source MAC address. If you want, you can change this behavior with the **port-channel load-balance** command:

```
SW1#show EtherChannel load-balance
```

```
EtherChannel Load-Balancing Configuration:
```

```
src-mac
```

```
EtherChannel Load-Balancing Addresses Used Per-Protocol:
```

```
Non-IP: Source MAC address
```

```
IPv4: Source MAC address
```

```
IPv6: Source MAC address
```

```
SW1(config)#port-channel load-balance ?
```

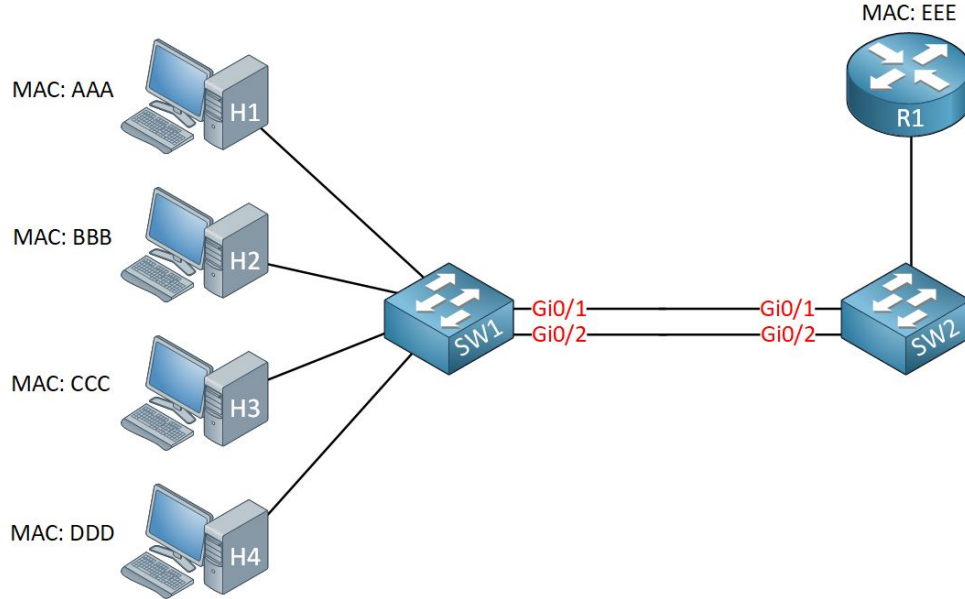
dst-ip	Dst IP Addr
dst-mac	Dst Mac Addr
src-dst-ip	Src XOR Dst IP Addr
src-dst-mac	Src XOR Dst Mac Addr
src-ip	Src IP Addr
src-mac	Src Mac Addr

EtherChannel Load Balancing



Strathmore
UNIVERSITY

- The options to choose from include combinations of source and/or destination MAC or IP addresses.
- Why load balancing?



SW1 has four computers. On the right side, SW2 is connected to a router. The default load-balancing mechanism is the source MAC address.

This means that ALL traffic from one MAC address will be sent down one and the same physical interface, for example:

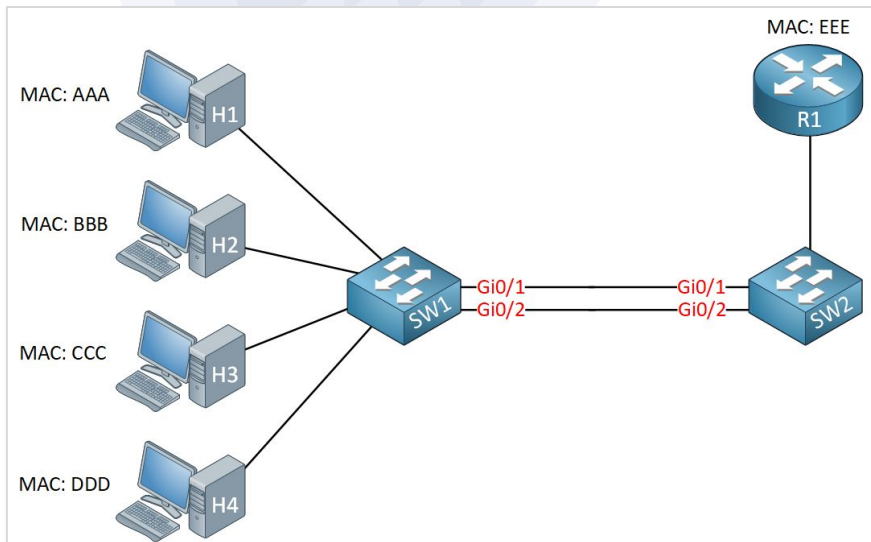
- MAC address AAA will be sent using SW1's gi0/1 interface.
- MAC address BBB will be sent using SW1's gi0/2 interface.
- MAC address CCC will be sent using SW1's gi0/1 interface.
- MAC address DDD will be sent using SW1's gi0/2 interface.

EtherChannel Load Balancing



Strathmore
UNIVERSITY

- Based on the graphic, both physical links on SW1 will be used for our EtherChannel, so depending on how much traffic the computers send, it will be close to a 1:1 ratio.
- As for SW2, since the connection is only to the router with MAC address EEEE. It will pick one of the physical interfaces so ALL traffic from the router will be sent down interface GigabitEthernet 0/1 or 0/2. One of the physical links won't be used at all.
- If this is the case, we should change our load-balancing algorithm.



- EtherChannels support multiple load-balancing options. Make sure you select a load-balancing algorithm that matches your traffic patterns.
- If we switch the load balancing to destination MAC address on SW2 then traffic from our router to the computers will be load-balanced amongst the different physical interfaces because we have multiple computers with different destination MAC addresses.
- The command is:

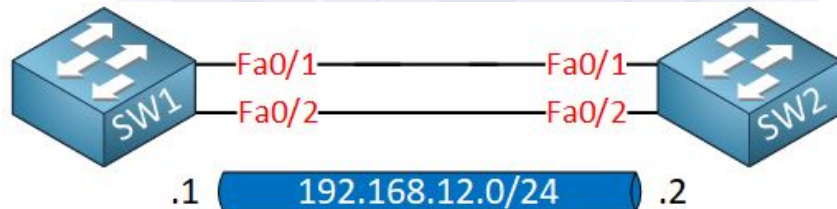
```
SW2(config)#port-channel load-balance dst-mac
```

Layer 3 EtherChannels



Strathmore
UNIVERSITY

- A layer three etherchannel is similar to an interface on a router. The switch won't "switch" traffic on this interface but route it. Because it's a layer three interface, we configure an IP address on it.
- Before configuring the port channel settings, you need to make sure that all interfaces have the exact same configuration.
- Once you use the channel-group command, the port-channel interface will automatically inherit all settings from your physical interface. If you forget to run the no switchport command on an interface, your etherchannel will be layer 2 instead of layer 3



```
SW1(config)#interface range fastEthernet 0/1 - 2
SW1(config-if-range)#no switchport
SW1(config-if-range)#channel-group 12 mode on
Creating a port-channel interface Port-channel 12
```

```
SW2(config)#interface range fa0/1 - 2
SW2(config-if-range)#no switchport
SW2(config-if-range)#channel-group 12 mode on
Creating a port-channel interface Port-channel 12
```

Layer 3 EtherChannels



- This creates the Etherchannel. We can verify the config through the **“Show etherchannel <ID> summary”**
- Port-channel 12 interface is layer 3, and it's operational. This can be assigned an IP address on this port-channel interface:

```
SW1(config)#interface port-channel 12
SW1(config-if)#ip address 192.168.12.1 255.255.255.0
```

```
SW2(config)#interface port-channel 12
SW2(config-if)#ip address 192.168.12.2 255.255.255.0
```

```
SW1#show etherchannel 12 summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

Group	Port-channel	Protocol	Ports
12	Po12(RU)	-	Fa0/1(P) Fa0/2(P)



Strathmore
UNIVERSITY